

Capability configuration, ambidexterity and performance: Evidence from service outsourcing sector

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ABSTRACT

How to resolve the dilemma of ambidexterity, or simultaneous alignment and adaption, is a vital issue in the operational field. Based on the resource management model, we argue that capability configurations could work as effective strategies for leveraging the potential of capabilities and benefit the pursuit of contradictory ambidexterity. This study explores how complementary and balance configurations of technological and marketing capabilities influence alignment and adaption and improve firm performance. Through regression analysis of data collected from service-outsourcing vendors in China, we provide evidence that both the complementary and balance configurations of these two capabilities can be used to achieve alignment and adaption, thereby improving firm performance. Moreover, whereas the complementary configuration of technological and marketing capabilities helps achieve alignment and adaption simultaneously, the balance configuration helps achieve only adaption. This study provides intriguing insights into resolving the dilemma of ambidexterity through exploiting the capability configuration strategy, which deserves scholars' attention in future studies.

1. Introduction

Ambidexterity comprising the simultaneous pursuit of alignment and adaption has attracted the growing attention of organizational theorists (for a recent literature review, see a study by O'Reilly and Tushman, 2013). At the operational level, ambidexterity (alignment and adaption) means that firms should not only produce outcomes that are efficiently aligned with objectives, but also be adaptive to demand changes in the task environment (Tiwana, 2008). Research has suggested that it is difficult for a firm to achieve alignment and adaption simultaneously due to their competing technological, operational and organizational requirements (Gibson and Birkinshaw, 2004; Tiwana, 2008). Nevertheless, scholars have argued that reconciling alignment and adaption to a large degree lends an organization a long-term competitive advantage (Cao et al., 2009; He and Wong, 2004; Menguc and Auh, 2008).

An organization's capability resides in and is exercised through its processes and routines (Song et al., 2005). As a way of integrating various capabilities, an organization's capability configuration essentially represents the coordination of the organization's different processes and routines (Teece et al., 1997; Eisenhardt and Martin, 2000; Teece, 2007). An effective capability configuration often displays high levels of coherence

and effectiveness and influences a firm's capacity to be aligned with operational objectives in terms of service quality, cost, delivery time and production lead time and to adapt to unexpected changes in timely responsiveness and flexibility (Teece, 2007). In this view, capability configurations help to benefit a firm's alignment and adaption jointly. However, researchers have only explored solutions to the dilemma of alignment and adaption from the perspectives of firm structure, context and leadership (Tushman et al., 1996; Ebben and Johnson, 2005; Gibson and Birkinshaw, 2004; Raisch and Birkinshaw, 2008) and have neglected the important role of capability configurations in achieving these two goals simultaneously. This limits our understanding of how to integrate a firm's internal capability of realizing ambidexterity in today's dynamic environment.

Capabilities are complex bundles of skills and accumulated knowledge embedded in organizational processes and routines and serve as critical sources of sustainable competitive advantage by enabling firms to coordinate activities and make use of assets (Day, 1994). Capabilities can be categorized by their functional areas such as operations, human management and information systems, which concerns firms' static ability to perform functional activities (Krasnikov and Jayachandran, 2008; Lun et al., 2016). The role of technological and marketing

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capabilities in driving superior firm performance has gained the attention and interest of many researchers (Song et al., 2005; Su et al., 2012). Although these studies have revealed that technological and marketing capabilities should be effectively coordinated in complementary configurations and deployed based on environmental conditions, the trade-off between technological and marketing capabilities has not received enough attention.

The trade-off between technological and marketing capabilities is a significant issue given the concerns surrounding capability coordination. Specifically, firms that rely only on their marketing capabilities may overlook technological opportunities and misallocate their technological resources. In comparison, an excessive technological capability may lead to a firm overlooking the voices of customers and impede its effective response to changes in demand (Hortinha et al., 2011). In this view, the trade-off between the two capabilities is unavoidable and thus requires more attention from researchers.

Based on the resource management model, this study builds an integrated framework to address these research gaps and investigate how the complementary and balance configurations of technological and marketing capabilities affect a firm's alignment and adaption practices at the operational level and enhance firm performance. This study attempts to address three research questions. First, how can a firm configure its technological and marketing capabilities? Second, should complementary or balance configurations of technological and marketing capabilities be beneficial to a firm's alignment or adaption or both? Third, how does a firm's capability configuration improve its performance through ambidexterity (alignment and adaption) practices? We attempt to answer these questions based on 250 samples taken from high-tech based service-outsourcing vendors in China.

This study makes three contributions to the literature. First, previous studies have focused on the complementary configuration of technological and marketing capabilities (Li et al., 2005; Song et al., 2005; Krasnikov and Jayachandran, 2008; Su et al., 2012). Based on these work, we further argue that balance configuration of technological and marketing capabilities is also a vital dimension that needs to be investigated. Because of the existence of core rigidities, a firm that overemphasizes one type of capability is likely to get involved in a competence trap. To avoid this trap, a firm needs to maintain the balance configuration across technological and marketing capabilities. Thus, exploring the roles of the balance configuration of technological and marketing capabilities would provide new insights into understanding the coordinative and synergic issues related to the various organizational capabilities.

Second, this study examines the driving forces of a firm's alignment and adaption practices from the perspective of the technological and marketing capability configurations. This would identify additional theoretical and managerial implications to consider when choosing a complementary or balanced strategy to integrate a firm's various capabilities and resolve its ambidexterity dilemma.

Third, this study provides valuable knowledge related to the merits of two kinds of capability configurations by modeling ambidexterity as the mediating link between a firm's capability configurations and performance. Although consensus has been reached about how capability configurations work as critical sources of comparative advantage and superior firm performance, the mediating roles between capability coordination and firm performance have not been explored. Thus, this study of the mediating role of ambidexterity between capability configurations and firm performance at the operational level deserves the further attention of scholars.

2. Theory background

2.1. Capabilities and capability configurations

Capabilities are embedded in an organization and its processes and cannot easily be transferred from one firm to another (Barney, 2001). Capabilities refer to a firm's capacity to deploy resources and can be

considered as special or high-order nontransferable resources that enhance the productivity of other ordinary resources (Makadok, 2001). These resources are tangible or intangible assets such as geographic location, factory equipment, a superior sales force and intellectual property (Mahoney and Pandian, 1992; Wernerfelt, 1995; Makadok, 2001). Strategic research has indicated that possessing rare, valuable, durable and inimitable resources and capabilities offers a firm a comparative advantage (Barney, 1995; Priem and Butler, 2001; Lambe et al., 2002).

Studies based on the resource management model have suggested that merely possessing resources and capabilities does not guarantee the development of competitive advantages. There is a need for the resource and capability configurations that help firms to achieve a long-term comparative advantage (Sirmon and Hitt, 2003; Sirmon et al., 2007). The configuration of idiosyncratic capabilities reconfigures those capabilities, decreases resource deficiency and generates new resource applications, creating a temporary competitive edge (Kogut and Zander, 1992; Teece et al., 1997; DeSarbo et al., 2005). To maximize the value of their capabilities, firms should integrate them in a more efficient way. Scholars have addressed capability configurations such as the complementary configurations of knowledge management and IT capabilities (Mao et al., 2014), marketing and innovation capabilities (Ngo and O'Cass, 2012) and technological and marketing capabilities (Song et al., 2005; Song and Thieme, 2006; Krasnikov and Jayachandran, 2008; Prašnikar et al., 2008).

Capabilities can be classified as inside-out and outside-in capabilities depending on the orientation and focus of the defining process (Day, 1994). As a kind of inside-out capability, technological capability refers to a firm's ability to deploy various technological resources (Afuah, 2002), which allows a firm to exploit important possibilities (Day, 1994). It is the internal capability of a firm to develop and produce new technologies that help the firm respond to the changing technological environment. Marketing capability refers to a firm's ability to build links with its customers (Day, 1994; Song et al., 2005). It is a kind of outside-in capability that enables a firm to understand its customers and create closer relationships with its customers, channel members and suppliers. With marketing capability, a firm can sense outside opportunities or possibilities and decide how best to serve potential customers. Because the effective configuration of capabilities contributes more to a firm's comparative advantage than simply having capabilities (Su et al., 2012), a firm must integrate its technological and marketing capabilities into a complementary configuration, and engage in high levels of both technological and marketing capabilities simultaneously to leverage their complementarities in order to enhance a firm's comparative advantage.

However, a firm's core capability is rigid. If a firm overemphasizes one kind of capability that strategically differentiates it from its other competitors, it may become reluctant to adapt to changes in the environment (Leonard-Barton, 1992; Levinthal and March 1993; Tushman et al., 1996; Schreyögg and Kliesch-Eberl, 2007; Lin et al., 2013). This flipside of a firm's core capability may hinder its long-term viability. Thus, there is a need for a firm to resolve trade-off issues when it hopes to integrate two different capabilities into a capability configuration. An effective capability configuration may need to make appropriate trade-offs and balances. For example, Kor and Mesko (2013) argued that CEOs must consider the trade-offs and try to create a proper balance between generic management capabilities and specialized management capabilities to enhance the executive team's absorptive capacity. Zhan and Chen (2013) suggested that managing an appropriate balance between exploration and exploitation capabilities is critical for firm survival and prosperity.

When a firm overemphasizes its strong technological capability, it may come to rely on its knowledge base, produce a new product or service that is consistent with past organization routines and technological trajectories and thus fail to satisfy consumers' evolving needs (Lavie, 2006; Zhou and Wu, 2010). In comparison, when a firm overemphasizes its marketing capability, it may increase its risk of becoming stuck

developing incremental innovations based on customer feedback. Thus, it is imperative for a firm to find a balance point between its technological and marketing capabilities in order to optimally distribute resources. Although the complementary perspective dominates the researches of capability configuration, only the complementary perspective can't offer the whole picture of the reality without describing the balance aspects of capability configuration. As such, both the complementary and balance capability configurations should all be addressed.

2.2. The mediating role of ambidexterity

As complex bundles of skills and accumulated knowledge, capabilities enable firms to deploy their resources (inputs) and achieve desired ends. Some scholars have argued that capabilities allow a firm to acquire and deploy resources to facilitate the production and delivery of goods or services, thereby benefitting a firm's internal operations (Rungtusanatham et al., 2003; Palvia et al., 2010; Shou et al., 2014). For example, technological capability can help a firm exploit new external technologies to improve its management system and production process, leading to low-cost, efficient and consistent delivery and maximizing system flexibility. Marketing capability allows a firm to integrate resources, information, knowledge or know-how from external customers and suppliers into its internal operations, thereby facilitating the input of high-quality materials and helping the firm improve the responsiveness of its internal operations by predicting customers' evolving needs (Rungtusanatham et al., 2003; Shou et al., 2014). Leveraging a firm's technological and marketing capabilities creates several internal operational benefits. It improves a firm's alignment by increasing its operational effectiveness, quality and speed and lowering its costs, and improves a firm's adaption by making its system more flexible and allowing it to provide more options and product customizations.

Alignment, which is associated with efficiency, means that a firm should achieve its expected technical, functional and customer requirements while meeting its scheduling and budget goals. Adaption, which is associated with flexibility, requires a firm to meet a variety of customer expectations and address the uncertainty involved by increasing its range of available products or adjusting its output volume quickly without experiencing excessive cost or time increases, organizational disruptions or firm losses. These two operational goals in the service delivery process are both related to ambidexterity (Gibson and Birkinshaw, 2004). Since the alignment has the potential for lowering price while the adaption can offer greater responsiveness to the specific product or delivery service needs of customer (Ebben and Johnson, 2005; Swink et al., 2005; Eisenhardt et al., 2010), both of them are considered to be important contributors to a firm's performance.

3. Hypothesis

3.1. Complementary configuration and ambidexterity

To achieve alignment, a firm should fulfill its clients' requirements efficiently by meeting its project objectives on schedule and delivering all desirable features and functionalities within budget. In this study, we argue that both marketing and technological capabilities can improve alignment interactively. The sensing component of marketing capability allows a firm to monitor its competitors' situation and developing trends, which help the firm identify operational weaknesses or defects. Meanwhile, technological capability allows a firm to provide technological solutions to remedy a firm's weaknesses or defects and hence ensure efficient operations that fulfill task requirements at a low cost and within an expected time range, leading to an improvement in alignment. Furthermore, the bonding component of marketing capability allows a firm to integrate the activities of customers, suppliers and channel members into its operations, providing the firm with accurate customer demand information and qualified inputs that help the firm achieve

alignment (Day, 1994; Prašnikar et al., 2008; Yu et al., 2014). In the meantime, technological capability can teach firms to transform customer demand information and qualified inputs into desired outcomes and help them offer services or products with desirable characteristics based on consumers' requirements (Afuah, 2002; Ortega, 2010; Wu, 2014).

To achieve adaption, a firm should be able to accommodate unexpected changes in a timely manner during project execution. These changes mainly result from customers' evolving needs, including calling for a different scope and volume of products or services, adjusting the delivery process and resolving unexpected problems. The sensing component of marketing capability can help a firm regularly anticipate customers' requirements by perceiving market trends and events (Krasnikov and Jayachandran, 2008; Merrilees et al., 2011). This may activate an operational adjustment plan to adapt to unexpected changes, but must be supported by the firm's technological capability. And the bonding component of marketing capability can help a firm involve customers in the joint problem-solving process and ensure a commitment to receiving uninterrupted high-quality inputs from suppliers. This improves a firm's ability to adapt to changes in the midst of demand fluctuations. Meanwhile, technological capability can facilitate such coordination between customers and other supply chain members by introducing new processes and information technologies. Based on the preceding analysis, we make the following hypotheses.

Hypothesis 1a. The complementary configuration between technological and marketing capabilities is positively related to alignment.

Hypothesis 1b. The complementary configuration between technological and marketing capabilities is positively related to adaption.

3.2. Balance configuration and ambidexterity

The balance configuration of technological and marketing capabilities implies that the weights of the two capabilities are more or less equal. We posit that balance reflects a firm's dominant logic: better balance means that neither technological nor marketing capability is dominant. We expect such domain logic to have a positive influence on alignment and adaption.

If a firm wants to achieve alignment, excessive technological or marketing capabilities will trap the firm in a dominant logic that makes it overly dependent on one capability, thereby impeding the synergistic effect of the two capabilities on the firm's alignment. Marketing capability can help a firm identify its operational weaknesses compared with its competitors and facilitates the inflow of high-quality inputs from supply chain members, helping the firm achieve alignment (Prašnikar et al., 2008; Mollenkopf et al., 2011). However, the lack of use of a firm's technological capability makes it more difficult to optimize internal operations, fix operational defects and coordinate supply chain members, thereby hindering the effect of marketing capability on alignment. In contrast, overemphasizing the benefits of technological capability inspires pure technical thinking to achieve alignment. This makes firms ignore that market trends provide opportunities for operational improvement and establish coordinative and cooperative relationships between supply chain members.

Excessive technological capability similarly leads a firm seeking to achieve adaption to overlook market information, the voice of the customer and efforts to solve problems jointly with customers and establish bonds with suppliers (Hortinha et al., 2011), which affect the firm when the needs of its customers change. In contrast, excessive marketing capability is futile for a firm, as without the support of a firm's strong technological capability, such as adjusting its internal operational system to prepare for fluctuations in customer demand and coordinating supply chain members while using new processes and information technologies to deal with unexpected problems or changes, no marketing scheme can be implemented properly. Thus, we posit that firms should find a balance point between these two capabilities rather than

overemphasize one capability to enhance their alignment and adaption.

Hypothesis 2a. The balance configuration of technological and marketing capabilities is positively related to alignment.

Hypothesis 2b. The balance configuration of technological and marketing capabilities is positively related to adaption.

3.3. Ambidexterity and firm performance

Alignment requires firms to provide products or services of a prescribed quality over a predictable timeframe and budget. This requires firms to be efficient in their service delivery processes. Firm efficiency relates to how a firm maximizes its output from given inputs and how it minimizes inputs given a level of output. These factors reflect the firm's effective use of resources that avoid the unnecessary costs caused by misallocation (Ebben and Johnson, 2005; Swink et al., 2005; Eisenhardt et al., 2010). Achieving alignment enables a firm to acquire a low-cost comparative advantage. Such a comparative advantage is thought to promote better profitability, build market share through the potential for lower prices and allow a firm to adjust prices dynamically in response to market changes.

Hypothesis 3. Alignment is positively related to firm performance.

Adaption requires firms to adapt to unexpected changes in time. Firms that achieve adaption are more flexible in the face of project- or task-related changes. Research has indicated that a flexible firm can acquire comparative advantages in task-uncertain environments, leading to superior firm performance. A flexible firm is more able to adapt its operational processes and hence allow movement between different product designs and modifications in response to clients' volatile demands (Boyer and Lewis, 2002; Ebben and Johnson, 2005; Eisenhardt et al., 2010). In addition, a flexible firm is able to provide a wide range of products and volumes in response to unpredictable demand changes (Anand and Ward, 2004; Oke, 2013). Thus, achieving adaption allows a firm to cope with volatile and unpredictable demands considered to threaten its performance. Based on these analyses, we make the following Hypothesis.

Hypothesis 4. Adaption is positively related to firm performance.

3.4. Complementary configurations and firm performance

Complementary capabilities infer that a high degree of one capability can improve the benefits of another and vice versa (DeSarbo et al., 2005). Based on previous studies, integrating technological and marketing capabilities in a complementary configuration increases the combined magnitude of both capabilities, thereby improving firm performance (Song et al., 2005; Su et al., 2012). The integration of idiosyncratic capabilities helps to reconfigure capabilities, decrease resource deficiency and generate new resource applications (Kogut and Zander, 1992; Teece et al., 1997; DeSarbo et al., 2005; Su et al., 2012). Marketing capabilities are deemed as "know-how" routines for deploying market resources and are associated with marketing functions (Danneels, 2007). They always relate to marketing mix processes such as products; pricing; channel management; and marketing communications, planning and implementation (Song et al., 2005; Morgan et al., 2009). These marketing mix activities establish relationships with customers, suppliers and channel members and help firms acquire external information and identify potential opportunities for improving current services/products or developing new ones. Technological capability can allow a firm to effectively implement these improvement or development processes by inventing new technical knowledge and converting it to design or develop superior products and services (Cohen and Levinthal, 1990; Afuah, 2002; Danneels, 2007; Su et al., 2012). Thus, integrating such two capabilities in a complementary configuration can improve firm performance. Based on these analyses, we make the following Hypothesis.

Hypothesis 5. The complementary configuration between technological and marketing capabilities is positively related to firm performance.

3.5. Balance configurations and firm performance

Balance configurations of technological and marketing capabilities, or closer matches in the relative magnitudes of these two capabilities, contribute to firm performance by controlling performance risks. More specifically, a technology-oriented firm that focuses more on technological capability than marketing capability always engages in exploitive activities. A firm that accumulates knowledge in the technology domain can competently assimilate external information in familiar fields (Levinthal and March 1993; Montgomery and Lieberman, 1998; Zhou and Wu, 2010). A firm's self-reinforcing nature and organizational inertia encourage it to rely on its current knowledge base and produce new products or services that are consistent with past organizational routines and technology trajectories (Leonard-Barton, 1992; Lavie, 2006; Kristal et al., 2010). The firm may run the risk of failing to satisfy customers' evolving needs.

In contrast, when a firm overemphasizes its marketing capability, it increases its risk of neglecting opportunities to leverage new technologies. A firm with a strong marketing capability is likely to become too focused on its customers and competitors' voices when segmenting the market, developing new products and implementing marketing plans, and may thus overlook new technological progress and opportunities (Hortinha et al., 2011; Morgan, 2012). Consequently, a firm may be outperformed by its competitors that have discovered new markets supported by recently developed technology. Consistent with the foregoing logic, we make the following Hypothesis.

Hypothesis 6. The balance configuration of technological and marketing capabilities is positively related to firm performance.

4. Methods

4.1. Sampling and data collection

We developed a questionnaire in accordance with Gerbing and Anderson's (1988) recommended procedures. We created an English version after reviewing a substantial amount of literature. Following this, we used two independent translators to translate the English questionnaire into its Chinese equivalent. We then conducted a Chinese-to-English translation process to ensure conceptual equivalence. For the purpose of ensuring face and content validity, we conducted five in-depth interviews with two marketing professors and three senior managers. We then revised the questionnaire in accordance with their responses to help ensure its completeness, relevance and clarity. Following this, we conducted a pretest with a sample size of 30 senior firm managers. In addition to completing the questionnaire, these senior managers provided useful feedback about its overall design and wording. We then refined and finalized the questionnaire based on the results.

The samples comprised high-tech-based service-outsourcing firms in China. Technological and marketing capabilities are critical for the performance of high-tech-based firms and help these firms monitor market changes and respond quickly to acquire emerging opportunities in emerging economies such as China (Wang et al., 2004; Li et al., 2005). In the service offshore outsourcing context, vendors always face the alignment goal of providing services to clients in accordance with contractual terms, such as expected quality quantification, service performance, time and budget. Vendors are also asked to fulfill an adaption goal to meet clients' changing demands. Thus, the Chinese service-outsourcing industry provides an ideal setting in which to test the research model. We chose typical firms from 20 key cities such as Shanghai, Suzhou, Dalian, Xian, Beijing and Chongqing to cover eastern, middle and western China. We considered these 20 cities representative for two reasons. First, the State Council of China designated them model outsourcing cities

according to a 2009 service-outsourcing industry development project. Second, the cities had mature outsourcing zones/parks, where a great many vendors were proactively taking on outsourcing projects from clients. We sent invitations to the potential respondents while assuring them that all of their personal information would be kept strictly confidential. The trained members from a professional advisory firm in Beijing sent our questionnaire to the respondents in these select Chinese cities.

We obtained 250 usable responses from 625 firms, representing a 40% response rate. To overcome potential common method bias, we sent each firm two questionnaires to be completed by different top management team personnel, such as chief executive officers, general managers and chief operational managers. We included one item asking the respondents about their specific tenures in their current firm positions to further validate their qualifications. Most of the 250 vendors were small (82%) in size with 500 or fewer employees. This result suggests that the average respondent tenure was 5.33 years (see Table 1). In addition, using T-tests we found no significant differences for age, monthly disposable income, education and key constructs between the early respondents (responses received within the first month) and the late respondents (responses received within the second month or later), indicating no response bias.

4.2. Measures

We measured the constructs using 7-point multi-item scales drawn or adapted from the established literature.

4.2.1. Technological capability

We measured this construct using five items adopted from a study by Chen et al. (2009). We used these items to assess a firm's ability to exploit its various technological resources. We asked the respondents to indicate the extent to which they agreed or disagreed with the following statements (1 = “strongly disagree,” 7 = “strongly agree”): (1) we have high-profile technological background personnel on our founding team; (2) we have our own product or process patents and copyright; (3) we make large financial investments in R&D and product development; (4) we actively share the latest technology and know-how with our business partners; and (5) innovation is greatly emphasized within the firm.

4.2.2. Marketing capability

We adopted this measurement from a study by Ngo and O'Cass, 2012 and applied it using 7-point scales (1 = “much worse than competitors,” 7 = “much better than competitors”). The items included the following: (1) incorporation of customer needs into product and service marketing; (2) pricing program development; (3) distribution system development; (4) marketing communication program development; (5) marketing planning skills; and (6) marketing activity implementation.

4.2.3. Complementary configuration between technological and marketing capabilities (CD)

CD concerns a firm's combined magnitude of technological and marketing capabilities, which is also defined as attaining high levels of both these two kinds of capabilities. The measures for technological and marketing capabilities were adapted from previous researches as detailed above.

$$CD = \text{Score of technological capability} \times \text{Score of marketing capability}$$

Notes: scores of technological and marketing capability have been centered on their means in this formula.

4.2.4. Balance configuration between technological and marketing capabilities (BD)

BD relates to the balance, or relative magnitudes of technological and marketing capabilities. To operationalize BD, we use the absolute difference between these two kinds of capabilities, following He and Wong's, 2004 treatment. To facilitate interpretation, we reverse this measure by subtracting the difference score from 7 so that a higher value indicates greater BD.

$$BD = 7 - |\text{Score of technological capability} - \text{Score of marketing capability}|$$

4.2.5. Alignment and adaption

We adopted these two constructs as dimensions of ambidexterity from studies by Gibson and Birkinshaw (2004) and Tiwana, 2008. We asked the respondents to rate items related to vendor service performance using 7-point scales (1 = “strongly disagree,” 7 = “strongly agree”). The alignment items included the following: the service we provide to clients (1) is within budget; (2) is on schedule; (3) delivers all desirable features and functionality; (4) meets key project objectives and business needs; and (5) is very successful overall. The adaption items included the following: our company has been able to (1) successfully manage scope changes; (2) adjust the delivery process to meet clients' requirements; (3) meet clients' individual requirements; and (4) resolve unexpected problems.

4.2.6. Firm performance

We adopted this measurement from a study by Rai and Tang (2010). We asked the respondents to rate their firm performance relative to that of their principal competitors in the market using 7-point scales (1 = “much worse than competitors,” 7 = “much better than competitors”). The following items were included: (1) sales growth; (2) return on sales after taxing; (3) competitive position; (4) client resources; (5) market share; and (6) new market acquisitions.

4.2.7. Control variables

We used firm size as a controlled variable because larger organizations might have had access to more resources that enabled them to develop better markets and generate superior financial performance. We measured firm size as the logarithm of the number of employees. Firm age and the number of years a firm has provided outsourcing services reflect a vendor's experience or operational and service delivery reputations, which play an important role in driving firm performance based on previous research (Kumar et al., 2009). We therefore measured these control variables as the logarithm of the number of years in our study.

Table 1
Profile of companies.

Characteristics of companies	Profile
1. Firm age	%
Less than 3 years	2.4
3 to 6 years	33.2
6 to 9 years	23.6
9 to 12 years	21.2
More than 12 years	19.6
2. Location	%
Eastern China	86
Middle China	2.4
Western China	11.6
3. Number of employees	%
Less than 200	61.2
200 to 400	15.2
400 to 600	6
More than 600	17.6
5. Ownership of vendors	%
Listed enterprises	6.8
Private companies	41.2
State-owned companies	5.2
Joint ventures	15.6
Foreign-funded enterprises	22.8
Others	8.4
6. Services provided by sampled companies	%
Information technology outsourcing (ITO)	87.2
Business process outsourcing (BPO)	68
Knowledge process outsourcing (KPO)	44.8

4.3. Reliability and validity

We calculated Cronbach's alpha, the average variance extracted (AVE) and the composite reliability coefficients based on the purified and unidimensionality measures for each construct. As seen in Table 2, Cronbach's alpha ranged from 0.79 to 0.90, greater than the 0.7 threshold level. The composite reliability and AVE exceeded their threshold levels of 0.6 and 0.5, respectively. Furthermore, all of the factor loadings were highly significant at the 1% level. Together, the test results demonstrated adequate reliability and convergent validity.

We addressed the divergent validity in two ways. First, we ran a chi-square difference test using two-factor confirmatory measurement models with each possible pair of constructs (10 tests) to determine whether the restricted model (in which the correlation between the two constructs was fixed to 1) was significantly worse than the unrestricted

model (in which the correlation between the two constructs was estimated freely). All of the χ^2 differences between each of the restricted and unrestricted models were highly significant (e.g., technological and marketing capabilities: $\chi^2(1) = 207.82, p = 0.000$), and in every instance the restricted models showed a worse data fit, providing evidence of discriminant validity (Anderson and Gerbing, 1988). Second, following a recommendation made by Fornell and Larcker (1981) we verified that the AVE for each construct was greater than its shared variance with other constructs as represented by the square of its correlations with other constructs (see Table 2), supporting discriminant validity.

4.4. Common method bias test

Common method variance (CMV) may be a concern when a self-report questionnaire is used to collect data from the same respondents at the same time, especially if the dependent and independent variables are perceptual measures derived from the same respondents (Podsakoff and Organ, 1986). Failing to control for this would have caused systemic measurement errors that inflated or deflated observed relationships between the constructs in the study, generating type-one and type-two errors. Researchers have referred to this problem as common method bias, and recent studies have recommended some approaches to avoiding or correcting CMV (Chang et al., 2010). These approaches are categorized as ex ante and ex post, where the former are implemented in the design stage and the latter are implemented in the data analysis stage (Chang et al., 2010). We took the ex ante approach by assuring the respondents of their anonymity and confidentiality; informing the respondents that there were no right or wrong answers, but that they should answer honestly; and clarifying items to ensure that ambiguous and unfamiliar items were excluded. We also took the ex post approach by implementing the Harman one-factor test (Podsakoff and Organ, 1986). The results of un-rotated factor analysis based on the principle component method indicated that no one general factor accounted for the majority of the variance, suggesting that CMV was not significant.

5. Analysis and results

Table 3 shows the hierarchical regression results. As Models 2a and 5a show, the complementary configuration between technological and marketing capabilities significantly and positively influence alignment ($r = 0.185$; $p < 0.05$) and adaption ($r = 0.122$; $p < 0.05$), supporting Hypotheses 1a and 1b, respectively. The balance configuration of technological and marketing capabilities is positively related to adaption, as shown in Model 6a ($r = 0.160$; $p < 0.01$), but had no direct effect on alignment, as shown in Model 3a ($r = 0.101$; $p > 0.1$). Thus, Hypothesis 2b is supported and Hypothesis 2a is unsupported. Moreover, as shown in Table 3, both technological and marketing capabilities had a significant effect on alignment and adaption.

Table 4 summarizes the regression hierarchical results for Hypotheses 3–6. Model 2b tests Hypotheses 3 and 4, which assume that alignment and adaption are positively related to firm performance, respectively. The results support these two hypotheses. As shown in Models 3b and 4b, consistent with Hypotheses 5 and 6, both the complementary and balance configurations of technological and marketing capabilities have a significant effect on firm performance. Furthermore, Models 5b and 6b investigate the effect of the complementary and balance capability configurations on firm performance when controlling for alignment and adaption, respectively. Such effects become weaker compared with Models 3b and 4b, which do not control for alignment and adaption. This indicates that the value of the complementary capability configuration is translated to firm performance through alignment and adaption, and that the balance capability configuration is translated to firm performance only through adaption (as shown in Fig. 1).

Table 2
Measurement items.

	Cronbach's a	Factor loading	AVE	CR
Technological capability (adopted from a study by Chen et al. (2009))	0.79		0.56	0.86
(1) We have high-profile technological background personnel on our founding team.		0.76		
(2) We have our own product or process patents and copyright.		0.70		
(3) We make financial investments in R&D and product development.		0.74		
(4) We actively share the latest technology and know-how with our business partners.		0.75		
(5) Innovation is greatly emphasized within the firm.		0.78		
Marketing capability (adopted from a study by Ngo and O'Cass (2012))	0.90		0.68	0.93
(1) Incorporation of customer needs into product and service marketing.		0.77		
(2) Pricing program development.		0.77		
(3) Distribution system development.		0.84		
(4) Marketing communication program development.		0.83		
(5) Marketing planning skills.		0.86		
(6) Marketing activity implementation.		0.86		
Ambidexterity (adapted from a study by Tiwana (2008))				
Alignment: The service we provide to clients	0.79		0.54	0.86
(1) is within budget.		0.70		
(2) is on schedule.		0.73		
(3) delivers all desirable features and functionality.		0.76		
(4) meets key project objectives and business needs.		0.78		
(5) is very successful overall.		0.72		
Adaption: Our company has been able to	0.77		0.60	0.86
(1) successfully manage scope changes.		0.78		
(2) adjust the delivery process to meet clients' requirements.		0.79		
(3) meet clients' individual requirements.		0.80		
(4) resolve unexpected problems.		0.72		
Firm performance (adapted from a study by Rai and Tang (2010))	0.91		0.69	0.93
(1) Sales growth.		0.82		
(2) Return on sales after taxing.		0.79		
(3) Competitive position.		0.84		
(4) Client resources.		0.83		
(5) Market share.		0.85		
(6) New market acquisitions.		0.84		

Table 3

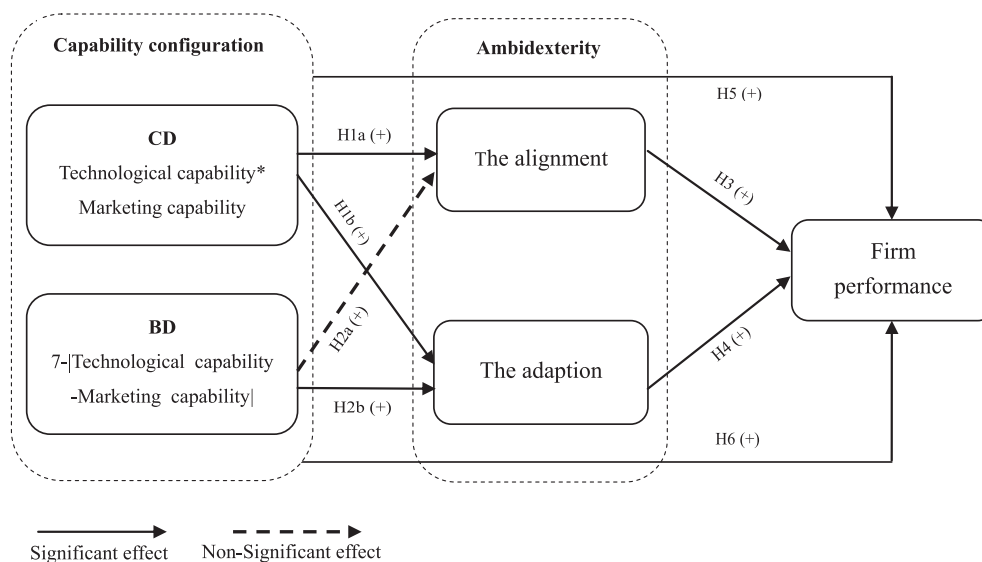
Results of hierarchical analyses with alignment and adaption as the dependent variable, respectively.

Dependent variables/Model	Alignment			Adaption		
	Model 1a	Model 2a	Model3a	Model 4a	Model 5a	Model 6a
Control variables						
Firm size	−0.08	−0.08	−0.06	−0.17**	−0.16	−0.16
Years of providing outsourcing services	0.14*	0.16*	0.14	0.11	0.11	0.10
Firm age	−0.22	−0.22***	−0.22***	−0.17	−0.11	−0.12
Explanatory variables						
Technological capability	0.30***	0.33***	0.31***	0.25***	0.24***	0.23***
Marketing capability	0.39***	0.39***	0.34***	0.28***	0.24***	0.33***
CD (Technological capability*Marketing capability)		0.18**			0.12**	
BD (7- Technological capability-Marketing capability)			0.10			0.16***
R-square	0.33	0.34	0.33	0.20	0.26	0.209
No. Obs.	250	250	250	250	250	250

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.**Table 4**

Results of hierarchical analyses with firm performance as the dependent variable.

Dependent variables/Model	Firm performance					
	Model 1b	Model 2b	Model 3b	Model 4b	Model 5b	Model 6b
Control variables						
Firm size	−0.03	−0.05	−0.03	−0.03	−0.06	−0.06
Years of providing outsourcing services	−0.05	−0.08	−0.05	−0.04	−0.09	−0.06
Firm age	0.27*	0.30***	0.27***	0.26***	0.31***	0.30***
Explanatory variables						
Technological capability	0.17*	0.14**	0.22**	0.19**	0.15**	0.16*
Marketing capability	0.39*	0.35***	0.39***	0.35***	0.35***	0.32***
Alignment		0.21***			0.21***	0.21***
Adaption		0.23***			0.23***	0.23***
CD (Technological capability*Marketing capability)			0.14***		−0.06	
BD (7- Technological capability-Marketing capability)				0.09**		0.09*
R-square	0.31	0.35	0.31	0.36	0.35	0.35
No. Obs.	250	250	250	250	250	250

* $p < 0.10$.** $p < 0.05$.*** $p < 0.01$.**Fig. 1.** Hypothesized model and empirical results.

6. Discussion

6.1. Theoretical contributions

This study investigates how technological and marketing capability configurations improve the ambidexterity of alignment and adaption and enhance firm performance. The results of empirical hierarchical regression analysis of 250 vendors in China indicates that the technological and marketing capability configurations could be important ways to attain alignment and adaption simultaneously and to realize the value created by the potential of capabilities in driving superior firm performance.

This study provides the following important theoretical contributions. First, it contributes to the research related to the resource management model by introducing the balance configuration of technological and marketing capabilities. According to the resource management model, capabilities should be coordinated as configurations that leverage and drive superior firm performance (Sirmon and Hitt, 2003; Sirmon et al., 2007). Consistent with this view, a number of empirical studies suggested that integrating various capabilities into the complementary configuration is an efficient way to realize the performance implication of capabilities by exploiting their synergies (Song et al., 2005; Song and Thieme, 2006; Krasnikov and Jayachandran, 2008; Prašnikar et al., 2008; Ngo and O'Cass, 2012; Su et al., 2012; Mao et al., 2014). However, the complementary configuration cannot resolve the trade-off issue of integrating capabilities. For example, overemphasizing technological capability may place a firm at risk of failing to satisfy customers' evolving needs, while overemphasizing marketing capability may increase a firm's risk of neglecting opportunities to leverage new technologies, both of which will harm firm performance (Hortinha et al., 2011). We provide evidence that with the balance configuration of these two capabilities, technological capabilities and marketing capabilities can compensate for each other's limitations or deficiencies. Therefore, the balance configurations of these two capabilities enable firms to avoid the competence trap and solve the trade-off issue, which improves firm performance. In doing so, this study contributes to the literature on capability configurations and sheds new light on how firms can increase performance by integrating capabilities in different ways.

Second, this study contributes to the ambidexterity theory literature though providing an intriguing insight into how to achieve ambidexterity from the perspective of capability configuration. According to previous studies, achieving ambidexterity presents a dilemma for firms because the two distinct dimensions of ambidexterity, i.e., alignment and adaption, compete with each other for resources (Gibson and Birkinshaw, 2004; Tiwana, 2008; O'Reilly and Tushman, 2011; Turner et al., 2012). Scholars have related ambidexterity (alignment and adaption) to a firm's structures, context and top management team characteristics, suggesting that a dual organization structure, autonomy context and team leaders who adopt paradoxical thinking are beneficial to achieving ambidexterity (Tushman et al., 1996; Ebben and Johnson, 2005; Gibson and Birkinshaw, 2004; Raisch and Birkinshaw, 2008; Turkuainen and Ruuska, 2013). However, as Chandrasekaran et al., 2012 observe, researchers have seldom provided answers to the question of what organizations must do to achieve ambidexterity from the perspective of leveraging capability. In this study, we investigate the effects of capability configuration on alignment and adaption separately. Our empirical results suggest that the complementary configuration of technological and marketing capabilities is beneficial for achieving both alignment and adaption simultaneously and that the balance configuration is beneficial for achieving only adaption. These findings provide solutions for resolving such a dilemma, and clearly indicate that complementarity is more likely than balance to provide an effective solution to resolve the ambidexterity dilemma.

Third, this study reveals two effective ways of integrating technological and marketing capabilities into configurations and investigates their underlying mechanisms that drive firm performance by introducing alignment and adaption as mediating variables. Although previous

research grounded in the resource-based view (RBV) and resource management model has demonstrated a positive relationship between the complementary configuration of these two capabilities and firm performance (Song et al., 2005; Song and Thieme, 2006), its underlying mechanism and balance configuration mechanism in driving firm performance remain in question. This study traces the implications of these issues by investigating the mediating role of ambidexterity in the relationship between capability configuration and firm performance. Consistent with the process perspective, which states that a related set of sequential variables leads to final outcomes (Markus and Robey, 1988), we argue that firm performance (output) is directly related to capability configuration (input) and that there are mediating process variables involved that signify a developmental progression. In this study, we observe both alignment and adaption to have significant partial mediating effects on the relationship between complementary configurations and firm performance. We also prove that adaption has one significant partial mediating effect on the relationship between the balance capability configuration and firm performance, and reveal a significant direct effect of the balance capability configuration on firm performance. As such, this study opens up the "black box" of the relationship between capability configurations and firm performance through introducing operational variables. Furthermore, our findings also respond to the argument of He and Wong (2004) that "a key challenge in isolating the impact of strategy variables on firm performance is to identify appropriate intermediary organizational or operational performance variables through which the impact of strategy variables is transmitted."

6.2. Managerial implication

This study has the following important managerial implications for firms in the business-to-business service industry.

First, our findings suggest that firms should integrate their technological and marketing capabilities in complementary or balance configurations to realize the potential of those capabilities in driving firm performance. In the process of developing new products or services, firms should make use of their technological capabilities to exploit new technologies such as cloud computing, mobile Internet technology and new media technology. Meanwhile, firms should leverage their market capabilities to understand their clients' evolving demands and market trends to successfully market new products and services. By leveraging these two capabilities as complementary configurations, firms may benefit more from new products and services in driving superior performance than they would from exploiting their capabilities separately. Furthermore, due to the positive effect of the balance configuration of technological and marketing capabilities on firm performance, our findings suggest that firms should try hard to find a balance point in leveraging these two capabilities. Firm managers should always be wary of becoming trapped in core rigidities due to an overemphasis on either technology or marketing capabilities.

Second, managers should resolve the dilemma posed by pursuing both alignment and adaption by integrating their capabilities into configurations. Studies have proposed the two most common approaches, i.e., structure and context, to guide firms in resolving such dilemmas (Gibson and Birkinshaw, 2004; Raisch and Birkinshaw, 2008). The structure approach achieves ambidexterity by dividing organization units into two parts that focus on alignment activities and adaption, respectively (Benner and Tushman, 2003). The context approach achieves ambidexterity by empowering employees to switch time between routine or alignment and nonroutine or adaption tasks (Gibson and Birkinshaw, 2004). This study suggests that it is more important for managers to resolve such dilemmas from a capability configuration perspective. If a firm wants to achieve both alignment and adaption simultaneously, its managers may leverage its complementary configurations. If the firm wants to achieve adaption, its managers may leverage its balance configurations. These findings provide managers with another way to achieve both alignment and adaption simultaneously.

6.3. Limitations and future study

This study has several limitations that should be explored in the future. First, we collected data from only one industry (the service-outsourcing industry) in China. Although this setting helps to eliminate industry-level variations that present potential noise in testing the model, it also limits the generalizability of the findings to other industries. Nevertheless, this does not mean that our key assumptions, theoretical model and empirical findings are inapplicable to other industries in which firms face the dilemma posed by pursuing both alignment and adaption. Future studies should collect data from other industries to validate our proposed model. Second, this study explores the effect of technological and marketing capability configurations. Future studies may deepen our understanding of capability configurations by introducing other capabilities. Third, there is a contradiction between the complementary configuration and the balance configuration, in that seeking to increase the complementary configuration may decrease balance configuration and vice versa, which will make it difficult for a firm to make a decision. Thus, there is a need for future research to explore the contingent factors that influence the effects of the complementary configuration and the balance configuration. Such research could shed light on the type of capability configurations that should be prioritized by firms when they encounter this contradiction. Fourth, as shown in our analysis, alignment and adaption partially mediate the relationship between complementary and balance capability configurations and firm performance. There may be other mechanisms that mediate this relationship, and these also deserve scholars' attention. For instance, future studies may consider how the values of capability configurations are translated to firm performance through the development of new products or services. Lastly, although determining how to achieve ambidexterity has been an important research question since the 1990s, discussions about the topic have been insufficient until now. Future studies may seek to provide answers to this question at different levels and from other strategic perspectives.

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